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Simulation of device removal

Abstract

In one aspect, a computing device is configured to, while operating as a control device of a networked media playback system that includes a first set of network devices and a second set of network devices, (i) transmit a first message to each network device in the first set of network devices, wherein the first message comprises a command to, at a first time, temporarily disconnect from the computing device via a particular communication protocol, (ii) transmit a second message to each network device in the second set of network devices, wherein the second message comprises a command for the network device in the second set of network devices to, at a second time, test connectivity with the computing device; and (iii) after the second time, determine whether each network device in the second set of network devices was connected to the computing device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority under 35 U.S.C. § 120 as a continuation of U.S. Nonprovisional application Ser. No. 17/342,390, filed on Jun. 8, 2021, and titled "Simulation of Device Removal," which claims the benefit of priority under 35 U.S.C. § 119 to U.S. Provisional Patent App. No. 63/036,032, filed on Jun. 8, 2020, and titled "Simulation of Device Removal," the contents of each of which are incorporated herein by reference in their entirety.

FIELD OF THE DISCLOSURE

(1) The present technology relates to consumer goods and, more particularly, to methods, systems, products, features, services, and other elements directed to media playback systems or some aspect thereof.

BACKGROUND

(2) Options for accessing and listening to digital audio in an out-loud setting were limited until in

2003, when SONOS, Inc. filed for one of its first patent applications, entitled “Method for Synchronizing Audio Playback between Multiple Networked Devices,” and began offering a media playback system for sale in 2005. The SONOS Wireless HiFi System enables people to experience music from many sources via one or more networked playback devices. Through a software control application installed on a smartphone, tablet, or computer, one can play what he or she wants in any room that has a networked playback device. Additionally, using a controller, for example, different songs can be streamed to each room that has a playback device, rooms can be grouped together for synchronous playback, or the same song can be heard in all rooms synchronously.

(3) Given the ever-growing interest in digital media, there continues to be a need to develop consumer-accessible technologies to further enhance the listening experience.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Features, aspects, and advantages of the presently disclosed technology may be better understood with regard to the following description, appended claims, and accompanying drawings.

(2) FIG. 1A is a partial cutaway view of an environment having a media playback system configured in accordance with aspects of the disclosed technology.

(3) FIG. 1B is a schematic diagram of the media playback system of FIG. 1A and one or more networks.

(4) FIG. 2A is a functional block diagram of an example playback device.

(5) FIG. 2B is an isometric diagram of an example housing of the playback device of FIG. 2A.

(6) FIGS. 3A-3E are diagrams showing example playback device configurations in accordance with aspects of the disclosure.

(7) FIG. 4A is a functional block diagram of an example controller device in accordance with aspects of the disclosure.

(8) FIGS. 4B and 4C are controller interfaces in accordance with aspects of the disclosure.

(9) FIG. 5 is a message flow diagram between an example user device, an example first set of network devices, and an example second set of network devices, in accordance with aspects of the disclosure.

(10) FIG. 6A is an example network topology in accordance with aspects of the disclosure.

(11) FIG. 6B is an example network topology in accordance with aspects of the disclosure.

(12) FIG. 6C is an example network topology in accordance with aspects of the disclosure.

(13) FIG. 6D is an example network topology in accordance with aspects of the disclosure.

(14) FIG. 7 is a flow diagram of an example method in accordance with aspects of the disclosure.

(15) The drawings are for purposes of illustrating example embodiments, but it should be understood that the inventions are not limited to the arrangements and instrumentality shown in the drawings. In the drawings, identical reference numbers identify at least generally similar elements. To facilitate the discussion of any particular element, the most significant digit or digits of any reference number refers to the Figure in which that element is first introduced. For example, element **103a** is first introduced and discussed with reference to FIG. 1A.

DETAILED DESCRIPTION

I. Overview

(16) Media playback systems may contain one or more playback devices (e.g., zone players) that are in communication with one another and may be configured to play back media or audio content in synchrony with one another. In these media playback systems, playback devices may be added over time, and the newer devices may have newer hardware that enables new functionality that might not be fully supportable by older playback devices having older hardware. For instance,

older playback devices may have a particular set of network communication capabilities such that a particular network protocol (e.g., via a hub or bridge device, via a direct ethernet connection, etc.) may be used for the older playback device to effectively communicate with the other playback devices, controller devices, and/or audio content sources. For instance, the older playback devices may use a Spanning Tree Protocol (STP) via a root networking device in order to communicate with the other playback devices, controller devices, and/or audio content sources. On the other hand, a newer playback device may have a newer set of network communication capabilities such that a different network protocol (e.g., non-hierarchical mesh network) may be used for the newer playback device to effectively communicate with other playback devices, controller devices, and/or audio content sources.

(17) As a user adds newer playback devices to a media playback system of one or more pre-existing playback devices that uses one or more additional network devices (e.g., a root or bridge networking devices), the additional network devices may be rendered obsolete as the newer playback devices may provide the same functionality as the additional network devices. For instance, if the additional network devices include a bridge device and the user adds a newer playback device that can perform the same bridging function, then the bridge device is no longer necessary. Additionally, the additional network devices may be rendered obsolete because newer playback devices may not need the additional network devices in order to connect to the one or more pre-existing playback devices of the media playback system. For instance, using the previous example, if the bridge device serves to connect the playback devices to the one or more pre-existing playback devices of the media playback system and the user adds a newer playback device that connects directly to the media playback system, the bridge device is no longer necessary for the media playback system to operate. Further, the additional network devices may be rendered obsolete because the newer playback devices may not need the additional network devices in order to connect to a router or access point in order to retrieve for playback, audio content from a streaming music service. Continuing with the previous example, if the bridge device serves as a gateway for the media playback device to access streaming audio content via the broader Internet, and the newer playback device can access a router or access point directly without the bridge device, the bridge device may be no longer necessary.

(18) However, users may be unaware that the additional network devices are no longer necessary for the operation of their media playback system when they add the newer playback devices. One challenge presented with the removal of unnecessary hardware from a media playback system is that the user may not be able to easily determine whether removing one or more devices from the media playback system will disrupt the media playback system, rendering it unsuitable for the user's purposes. For example, a media playback system may include a bridge device that is configured to facilitate connection between the playback devices via the creation of a wireless mesh network. Over time, the user may add additional playback devices that connect to the media playback system through the wireless mesh network. However, the user may not fully appreciate the function the bridge device is performing and at a later time, may remove the bridge device, which in turn may render the media playback system inoperable. In this example, it may not be apparent to the user which devices (e.g., the bridge device) of the media playback system are necessary for continued operation.

(19) One potential solution to this challenge is for the user to manually disconnect every device that is connected to the media playback system and then systematically reconnect each disconnected device while testing connectivity to the media playback system. However, this process can be tedious and might not reliably identify the devices that can be removed from the system.

(20) To address this problem, a user device that is connected to the media playback system may simulate the removal of any additional network devices and/or older playback devices by temporarily disconnecting them from the remaining playback devices in the media playback system and subsequently testing the connection of the remaining playback devices to one another and the

user device. In this respect, the user device is able to, through the simulation, determine whether and which of the tested devices can be removed from the system without disrupting the connection of the other devices in the system. The user device may also present an identification of the devices that can be removed to the user for manual removal and/or disconnection.

(21) Example devices, systems, and methods configured in accordance with embodiments of this disclosure help address these and other challenges associated with simulating the removal of devices of a media playback system to identify devices that can be potentially removed from the media playback system without affecting the performance of the media playback system. In a first example embodiment, a computing device may include at least one processor, a network interface, and at least one non-transitory computer-readable medium with program instructions stored on the at least one non-transitory computer-readable medium that are executable by the at least one processor such that the computing device is configured to, while operating as a control device of a networked media playback system that includes a first set of network devices and a second set of network devices, (i) transmit, via the network interface, a first message to each network device in the first set of network devices, wherein the first message comprises a command for the network device in the first set of network devices to, at a first time, temporarily disconnect from the computing device via a particular communication protocol, (ii) transmit, via the network interface, a second message to each network device in the second set of network devices, wherein the second message comprises a command for the network device in the second set of network devices to, at a second time, test connectivity with the computing device, and (iii) after the second time, determine whether each network device in the second set of network devices is connected to the computing device.

(22) In some examples, a second example embodiment may include a non-transitory computer-readable medium with program instructions stored on the non-transitory computer-readable medium that are executable by the at least one processor such that the computing device is configured to perform the functions of the first example embodiment.

(23) In some examples, a third example embodiment may include a method of performing the functions of the first example embodiment.

(24) While some embodiments described herein may refer to functions performed by given actors, such as “users” and/or other entities, it should be understood that this description is for purposes of explanation only. The claims should not be interpreted to require action by any such example actor unless explicitly required by the language of the claims themselves.

II. Example Operating Environment

(25) FIGS. 1A and 1B illustrate an example configuration of a media playback system **100** (or “MPS **100**”) in which one or more embodiments disclosed herein may be implemented. Referring first to FIG. 1A, the MPS **100** as shown is associated with an example home environment having a plurality of rooms and spaces, which may be collectively referred to as a “home environment,” “smart home,” or “environment **101**.” The environment **101** comprises a household having several rooms, spaces, and/or playback zones, including a master bathroom **101a**, a master bedroom **101b** (referred to herein as “Nick's Room”), a second bedroom **101c**, a family room or den **101d**, an office **101e**, a living room **101f**, a dining room **101g**, a kitchen **101h**, and an outdoor patio **101i**. While certain embodiments and examples are described below in the context of a home environment, the technologies described herein may be implemented in other types of environments. In some embodiments, for example, the MPS **100** can be implemented in one or more commercial settings (e.g., a restaurant, mall, airport, hotel, a retail or other store), one or more vehicles (e.g., a sports utility vehicle, bus, car, a ship, a boat, an airplane), multiple environments (e.g., a combination of home and vehicle environments), and/or another suitable environment where multi-zone audio may be desirable.

(26) Within these rooms and spaces, the MPS **100** includes one or more computing devices. Referring to FIGS. 1A and 1B together, such computing devices can include playback devices **102**

(identified individually as playback devices **102a-102o**), network microphone devices **103** (identified individually as “NMDs” **103a-102i**), and controller devices **104a** and **104b** (collectively “controller devices **104**”). Referring to FIG. **1B**, the home environment may include additional and/or other computing devices, including local network devices, such as one or more smart illumination devices **108** (FIG. **1B**), a smart thermostat **110**, and a local computing device **105** (FIG. **1A**). In embodiments described below, one or more of the various playback devices **102** may be configured as portable playback devices, while others may be configured as stationary playback devices. For example, the headphones **102o** (FIG. **1B**) are a portable playback device, while the playback device **102d** on the bookcase may be a stationary device. As another example, the playback device **102c** on the Patio may be a battery-powered device, which may allow it to be transported to various areas within the environment **101**, and outside of the environment **101**, when it is not plugged in to a wall outlet or the like.

(27) With reference still to FIG. **1B**, the various playback, network microphone, and controller devices **104** and/or other network devices of the MPS **100** may be coupled to one another via point-to-point connections and/or over other connections, which may be wired and/or wireless, via a local network **111** that may include a network router **109**. For example, the playback device **102j** in the Den **101d** (FIG. **1A**), which may be designated as the “Left” device, may have a point-to-point connection with the playback device **102a**, which is also in the Den **101d** and may be designated as the “Right” device. In a related embodiment, the Left playback device **102j** may communicate with other network devices, such as the playback device **102b**, which may be designated as the “Front” device, via a point-to-point connection and/or other connections via the local network **111**. The local network **111** may be, for example, a network that interconnects one or more devices within a limited area (e.g., a residence, an office building, a car, an individual's workspace, etc.). The local network **111** may include, for example, one or more local area network (LANs) such as wireless local area networks (WLANs) (e.g., WI-FI networks, Z-WAVE networks, etc.) and/or one or more personal area networks (PANs) such as BLUETOOTH networks, wireless USB networks, ZIGBEE networks, and IRDA networks.

(28) As further shown in FIG. **1B**, the MPS **100** may be coupled to one or more remote computing devices **106** via a wide area network (“WAN”) **107**. In some embodiments, each remote computing device **106** may take the form of one or more cloud servers. The remote computing devices **106** may be configured to interact with computing devices in the environment **101** in various ways. For example, the remote computing devices **106** may be configured to facilitate streaming and/or controlling playback of media content, such as audio, in the home environment **101**.

(29) In some implementations, the various playback devices, NMDs, and/or controller devices **104** may be communicatively coupled to at least one remote computing device associated with a voice assistant service (“VAS”) and at least one remote computing device associated with a media content service (“MCS”). For instance, in the illustrated example of FIG. **1B**, remote computing devices **106a** are associated with a VAS **190** and remote computing devices **106b** are associated with an MCS **192**. Although only a single VAS **190** and a single MCS **192** are shown in the example of FIG. **1B** for purposes of clarity, the MPS **100** may be coupled to multiple, different VASes and/or MCSes. In some implementations, VASes may be operated by one or more of AMAZON, GOOGLE, APPLE, MICROSOFT, NUANCE, SONOS or other voice assistant providers. In some implementations, MCSes may be operated by one or more of SPOTIFY, PANDORA, AMAZON MUSIC, or other media content services.

(30) As further shown in FIG. **1B**, the remote computing devices **106** further include remote computing device **106c** configured to perform certain operations, such as remotely facilitating media playback functions, managing device and system status information, directing communications between the devices of the MPS **100** and one or multiple VASes and/or MCSes, among other operations. In one example, the remote computing devices **106c** provide cloud servers for one or more SONOS Wireless HiFi Systems.

(31) In various implementations, one or more of the playback devices **102** may take the form of or include an on-board (e.g., integrated) network microphone device. For example, the playback devices **102a-e** include or are otherwise equipped with corresponding NMDs **103a-e**, respectively. A playback device that includes or is equipped with an NMD may be referred to herein interchangeably as a playback device or an NMD unless indicated otherwise in the description. In some cases, one or more of the NMDs **103** may be a stand-alone device. For example, the NMDs **103f** and **103g** may be stand-alone devices. A stand-alone NMD may omit components and/or functionality that is typically included in a playback device, such as a speaker or related electronics. For instance, in such cases, a stand-alone NMD may not produce audio output or may produce limited audio output (e.g., relatively low-quality audio output).

(32) The various playback and network microphone devices **102** and **103** of the MPS **100** may each be associated with a unique name, which may be assigned to the respective devices by a user, such as during setup of one or more of these devices. For instance, as shown in the illustrated example of FIG. **1B**, a user may assign the name “Bookcase” to playback device **102d** because it is physically situated on a bookcase. Similarly, the NMD **103f** may be assigned the named “Island” because it is physically situated on an island countertop in the Kitchen **101h** (FIG. **1A**). Some playback devices may be assigned names according to a zone or room, such as the playback devices **102e**, **102l**, **102m**, and **102n**, which are named “Bedroom,” “Dining Room,” “Living Room,” and “Office,” respectively. Further, certain playback devices may have functionally descriptive names. For example, the playback devices **102a** and **102b** are assigned the names “Right” and “Front,” respectively, because these two devices are configured to provide specific audio channels during media playback in the zone of the Den **101d** (FIG. **1A**). The playback device **102c** in the Patio may be named portable because it is battery-powered and/or readily transportable to different areas of the environment **101**. Other naming conventions are possible.

(33) As discussed above, an NMD may detect and process sound from its environment, such as sound that includes background noise mixed with speech spoken by a person in the NMD's vicinity. For example, as sounds are detected by the NMD in the environment, the NMD may process the detected sound to determine if the sound includes speech that contains voice input intended for the NMD and ultimately a particular VAS. For example, the NMD may identify whether speech includes a wake word associated with a particular VAS.

(34) In the illustrated example of FIG. **1B**, the NMDs **103** are configured to interact with the VAS **190** over the local network **111** and/or the router **109**. Interactions with the VAS **190** may be initiated, for example, when an NMD identifies in the detected sound a potential wake word. The identification causes a wake-word event, which in turn causes the NMD to begin transmitting detected-sound data to the VAS **190**. In some implementations, the various local network devices **102-105** (FIG. **1A**) and/or remote computing devices **106c** of the MPS **100** may exchange various feedback, information, instructions, and/or related data with the remote computing devices associated with the selected VAS. Such exchanges may be related to or independent of transmitted messages containing voice inputs. In some embodiments, the remote computing device(s) and the media playback system **100** may exchange data via communication paths as described herein and/or using a metadata exchange channel as described in U.S. Patent Publication No. 2017-0242653 published Aug. 24, 2017, and titled “Voice Control of a Media Playback System,” which is herein incorporated by reference in its entirety.

(35) Upon receiving the stream of sound data, the VAS **190** determines if there is voice input in the streamed data from the NMD, and if so the VAS **190** will also determine an underlying intent in the voice input. The VAS **190** may next transmit a response back to the MPS **100**, which can include transmitting the response directly to the NMD that caused the wake-word event. The response is typically based on the intent that the VAS **190** determined was present in the voice input. As an example, in response to the VAS **190** receiving a voice input with an utterance to “Play Hey Jude by The Beatles,” the VAS **190** may determine that the underlying intent of the voice input is to

initiate playback and further determine that intent of the voice input is to play the particular song “Hey Jude.” After these determinations, the VAS **190** may transmit a command to a particular MCS **192** to retrieve content (i.e., the song “Hey Jude”), and that MCS **192**, in turn, provides (e.g., streams) this content directly to the MPS **100** or indirectly via the VAS **190**. In some implementations, the VAS **190** may transmit to the MPS **100** a command that causes the MPS **100** itself to retrieve the content from the MCS **192**.

(36) In certain implementations, NMDs may facilitate arbitration amongst one another when voice input is identified in speech detected by two or more NMDs located within proximity of one another. For example, the NMD-equipped playback device **102d** in the environment **101** (FIG. **1A**) is in relatively close proximity to the NMD-equipped Living Room playback device **102m**, and both devices **102d** and **102m** may at least sometimes detect the same sound. In such cases, this may require arbitration as to which device is ultimately responsible for providing detected-sound data to the remote VAS. Examples of arbitrating between NMDs may be found, for example, in previously referenced U.S. Patent Publication No. 2017-0242653.

(37) In certain implementations, an NMD may be assigned to, or otherwise associated with, a designated or default playback device that may not include an NMD. For example, the Island NMD **103f** in the Kitchen **101h** (FIG. **1A**) may be assigned to the Dining Room playback device **102l**, which is in relatively close proximity to the Island NMD **103f**. In practice, an NMD may direct an assigned playback device to play audio in response to a remote VAS receiving a voice input from the NMD to play the audio, which the NMD might have sent to the VAS in response to a user speaking a command to play a certain song, album, playlist, etc. Additional details regarding assigning NMDs and playback devices as designated or default devices may be found, for example, in previously referenced U.S. Patent Publication No. 2017-0242653.

(38) Further aspects relating to the different components of the example MPS **100** and how the different components may interact to provide a user with a media experience may be found in the following sections. While discussions herein may generally refer to the example MPS **100**, technologies described herein are not limited to applications within, among other things, the home environment described above. For instance, the technologies described herein may be useful in other home environment configurations comprising more or fewer of any of the playback, network microphone, and/or controller devices **104**. For example, the technologies herein may be utilized within an environment having a single playback device **102** and/or a single NMD **103**. In some examples of such cases, the local network **111** (FIG. **1B**) may be eliminated and the single playback device **102** and/or the single NMD **103** may communicate directly with the remote computing devices **106a—d**. In some embodiments, a telecommunication network (e.g., an LTE network, a 5G network, etc.) may communicate with the various playback, network microphone, and/or controller devices **104** independent of the local network **111**.

(39) While specific implementations of MPS's have been described above with respect to FIGS. **1A** and **1B**, there are numerous configurations of MPS's, including, but not limited to, those that do not interact with remote services, systems that do not include controllers, and/or any other configuration as appropriate to the requirements of a given application.

(40) a. Example Playback & Network Microphone Devices

(41) FIG. **2A** is a functional block diagram illustrating certain aspects of one of the playback devices **102** of the MPS **100** of FIGS. **1A** and **1B**. As shown, the playback device **102** includes various components, each of which is discussed in further detail below, and the various components of the playback device **102** may be operably coupled to one another via a system bus, communication network, or some other connection mechanism. In the illustrated example of FIG. **2A**, the playback device **102** may be referred to as an “NMD-equipped” playback device because it includes components that support the functionality of an NMD, such as one of the NMDs **103** shown in FIG. **1A**.

(42) As shown, the playback device **102** includes at least one processor **212**, which may be a clock-

driven computing component configured to process input data according to instructions stored in memory **213**. The memory **213** may be a tangible, non-transitory, computer-readable medium configured to store instructions that are executable by the processor **212**. For example, the memory **213** may be data storage that can be loaded with software code **214** that is executable by the processor **212** to achieve certain functions. The processor **212** may comprise any of a variety of processors including, for example, one or more general-purpose processors (GPPs) and/or one or more special-purpose processors (SPPs) (e.g., digital signal processors (DSPs), neural processors, etc.).

(43) In one example, these functions may involve the playback device **102** retrieving audio data from an audio source, which may be another playback device. In another example, the functions may involve the playback device **102** sending audio data, detected-sound data (e.g., corresponding to a voice input), and/or other information to another device on a network via at least one network interface **224**. In yet another example, the functions may involve the playback device **102** causing one or more other playback devices to synchronously playback audio with the playback device **102**. In yet a further example, the functions may involve the playback device **102** facilitating being paired or otherwise bonded with one or more other playback devices to create a multi-channel audio environment. Numerous other example functions are possible, some of which are discussed below.

(44) As just mentioned, certain functions may involve the playback device **102** synchronizing playback of audio content with one or more other playback devices. During synchronous playback, a listener may not perceive time-delay differences between playback of the audio content by the synchronized playback devices. U.S. Pat. No. 8,234,395 filed on Apr. 4, 2004, and titled “System and method for synchronizing operations among a plurality of independently clocked digital data processing devices,” which is hereby incorporated by reference in its entirety, provides in more detail some examples for audio playback synchronization among playback devices.

(45) To facilitate audio playback, the playback device **102** includes audio processing components **216** that are generally configured to process audio prior to the playback device **102** rendering the audio. In this respect, the audio processing components **216** may include one or more digital-to-analog converters (“DAC”), one or more audio preprocessing components, one or more audio enhancement components, one or more digital signal processors (“DSPs”), one or more buffers, and so on. In some implementations, one or more of the audio processing components **216** may be a subcomponent of the processor **212**. In operation, the audio processing components **216** receive analog and/or digital audio and process and/or otherwise intentionally alter the audio to produce audio signals for playback.

(46) The produced audio signals may then be provided to one or more audio amplifiers **217** for amplification and playback through one or more speakers **218** operably coupled to the amplifiers **217**. The audio amplifiers **217** may include components configured to amplify audio signals to a level for driving one or more of the speakers **218**.

(47) Each of the speakers **218** may include an individual transducer (e.g., a “driver”) or the speakers **218** may include a complete speaker system involving an enclosure with one or more drivers. A particular driver of a speaker **218** may include, for example, a subwoofer (e.g., for low frequencies), a mid-range driver (e.g., for middle frequencies), and/or a tweeter (e.g., for high frequencies). In some cases, a transducer may be driven by an individual corresponding audio amplifier of the audio amplifiers **217**. In some implementations, a playback device may not include the speakers **218**, but instead may include a speaker interface for connecting the playback device to external speakers. In certain embodiments, a playback device may include neither the speakers **218** nor the audio amplifiers **217**, but instead may include an audio interface (not shown) for connecting the playback device to an external audio amplifier or audio-visual receiver.

(48) In addition to producing audio signals for playback by the playback device **102**, the audio processing components **216** may be configured to process audio to be sent to one or more other

playback devices, via the network interface **224**, for playback. In example scenarios, audio content to be processed and/or played back by the playback device **102** may be received from an external source, such as via an audio line-in interface (e.g., an auto-detecting 3.5 mm audio line-in connection) of the playback device **102** (not shown) or via the network interface **224**, as described below.

(49) As shown, the at least one network interface **224**, may take the form of one or more wireless interfaces **225** and/or one or more wired interfaces **226**. A wireless interface may provide network interface functions for the playback device **102** to wirelessly communicate with other devices (e.g., other playback device(s), NMD(s), and/or controller device(s)) in accordance with a communication protocol (e.g., any wireless standard including IEEE 802.11a, 802.11b, 802.11g, 802.11n, 802.11ac, 802.11ad, 802.11af, 802.11ah, 802.11ai, 802.11aj, 802.11aq, 802.11ax, 802.11ay, 802.15, BLUETOOTH, 4G mobile communication standard, 5G mobile communication standard, and so on). A wired interface may provide network interface functions for the playback device **102** to communicate over a wired connection with other devices in accordance with a communication protocol (e.g., IEEE 802.3). While the network interface **224** shown in FIG. 2A includes both wired and wireless interfaces, the playback device **102** may in some implementations include only wireless interface(s) or only wired interface(s).

(50) In general, the network interface **224** facilitates data flow between the playback device **102** and one or more other devices on a data network. For instance, the playback device **102** may be configured to receive audio content over the data network from one or more other playback devices, network devices within a LAN, and/or audio content sources over a WAN, such as the Internet. In one example, the audio content and other signals transmitted and received by the playback device **102** may be transmitted in the form of digital packet data comprising an Internet Protocol (IP)-based source address and IP-based destination addresses. In such a case, the network interface **224** may be configured to parse the digital packet data such that the data destined for the playback device **102** is properly received and processed by the playback device **102**.

(51) As shown in FIG. 2A, the playback device **102** also includes voice processing components **220** that are operably coupled to one or more microphones **222**. The microphones **222** are configured to detect sound (i.e., acoustic waves) in the environment of the playback device **102**, which is then provided to the voice processing components **220**. More specifically, each microphone **222** is configured to detect sound and convert the sound into a digital or analog signal representative of the detected sound, which can then cause the voice processing component **220** to perform various functions based on the detected sound, as described in greater detail below. In one implementation, the microphones **222** are arranged as an array of microphones (e.g., an array of six microphones). In some implementations, the playback device **102** includes more than six microphones (e.g., eight microphones or twelve microphones) or fewer than six microphones (e.g., four microphones, two microphones, or a single microphones).

(52) In operation, the voice-processing components **220** are generally configured to detect and process sound received via the microphones **222**, identify potential voice input in the detected sound, and extract detected-sound data to enable a VAS, such as the VAS **190** (FIG. 1B), to process voice input identified in the detected-sound data. The voice processing components **220** may include one or more analog-to-digital converters, an acoustic echo canceller (“AEC”), a spatial processor (e.g., one or more multi-channel Wiener filters, one or more other filters, and/or one or more beam former components), one or more buffers (e.g., one or more circular buffers), one or more wake-word engines, one or more voice extractors, and/or one or more speech processing components (e.g., components configured to recognize a voice of a particular user or a particular set of users associated with a household), among other example voice processing components. In example implementations, the voice processing components **220** may include or otherwise take the form of one or more DSPs or one or more modules of a DSP. In this respect, certain voice processing components **220** may be configured with particular parameters (e.g., gain and/or

spectral parameters) that may be modified or otherwise tuned to achieve particular functions. In some implementations, one or more of the voice processing components **220** may be a subcomponent of the processor **212**.

(53) In some implementations, the voice-processing components **220** may detect and store a user's voice profile, which may be associated with a user account of the MPS **100**. For example, voice profiles may be stored as and/or compared to variables stored in a set of command information or data table. The voice profile may include aspects of the tone or frequency of a user's voice and/or other unique aspects of the user's voice, such as those described in previously-referenced U.S. Patent Publication No. 2017-0242653.

(54) As further shown in FIG. 2A, the playback device **102** also includes power components **227**. The power components **227** may include at least an external power source interface **228**, which may be coupled to a power source (not shown) via a power cable or the like that physically connects the playback device **102** to an electrical outlet or some other external power source. Other power components may include, for example, transformers, converters, and like components configured to format electrical power.

(55) In some implementations, the power components **227** of the playback device **102** may additionally include an internal power source **229** (e.g., one or more batteries) configured to power the playback device **102** without a physical connection to an external power source. When equipped with the internal power source **229**, the playback device **102** may operate independent of an external power source. In some such implementations, the external power source interface **228** may be configured to facilitate charging the internal power source **229**. As discussed before, a playback device comprising an internal power source may be referred to herein as a "portable playback device." Those portable playback devices that weigh no more than fifty ounces (e.g., between three ounces and fifty ounces, between five ounces and fifty ounces, between ten ounces and fifty ounces, between ten ounces and twenty-five ounces, etc.) may be referred to herein as an "ultra-portable playback device." Those playback devices that operate using an external power source instead of an internal power source may be referred to herein as a "stationary playback device," although such a device may in fact be moved around a home or other environment.

(56) The playback device **102** may further include a user interface **240** that may facilitate user interactions independent of or in conjunction with user interactions facilitated by one or more of the controller devices **104**. In various embodiments, the user interface **240** includes one or more physical buttons and/or supports graphical interfaces provided on touch sensitive screen(s) and/or surface(s), among other possibilities, for a user to directly provide input. The user interface **240** may further include one or more of lights (e.g., LEDs) and the speakers to provide visual and/or audio feedback to a user.

(57) As an illustrative example, FIG. 2B shows an example housing **230** of the playback device **102** that includes a user interface in the form of a control area **232** at a top portion **234** of the housing **230**. The control area **232** includes buttons **236a-c** for controlling audio playback, volume level, and other functions. The control area **232** also includes a button **236d** for toggling the microphones **222** to either an on state or an off state.

(58) As further shown in FIG. 2B, the control area **232** is at least partially surrounded by apertures formed in the top portion **234** of the housing **230** through which the microphones **222** (not visible in FIG. 2B) receive the sound in the environment of the playback device **102**. The microphones **222** may be arranged in various positions along and/or within the top portion **234** or other areas of the housing **230** so as to detect sound from one or more directions relative to the playback device **102**.

(59) As mentioned above, the playback device **102** may be constructed as a portable playback device, such as an ultra-portable playback device, that comprises an internal power source. In one embodiment, such a portable playback device may comprise a housing that includes a user interface in the form of a control area at a top portion of the housing. The control area may include a capacitive touch sensor for controlling audio playback, volume level, and other functions. The

housing of the portable playback device may be configured to engage with a dock that is connected to an external power source via a cable. The dock may be configured to provide power to the portable playback device to recharge an internal battery. In some embodiments, the dock may comprise a set of one or more conductive contacts positioned on the top of the dock that engage with conductive contacts on the bottom of the housing. In other embodiments, the dock may provide power from the cable to the portable playback device without the use of conductive contacts. For example, the dock may wirelessly charge the portable playback device via one or more inductive coils integrated into each of the dock and the portable playback device.

(60) In some embodiments, the playback device **102** may take the form of a wired and/or wireless headphone (e.g., an over-ear headphone, an on-ear headphone, or an in-ear headphone). For instance, in one such implementation, the playback device **102** may comprise a housing that includes a headband that couples a first earpiece to a second earpiece. Each of the earpieces may house any portion of the electronic components in the playback device, such as one or more speakers. Further, one or both of the earpieces may include a control area for controlling audio playback, volume level, and other functions. The control area may comprise any combination of the following: a capacitive touch sensor, a button, a switch, and a dial. The housing may further include first and second ear cushions that are coupled to the first and second earpieces, respectively. The ear cushions may provide a soft barrier between the head of a user and the earpieces to improve user comfort and/or provide acoustic isolation from the ambient (e.g., passive noise reduction (PNR)). In some implementations, the wired and/or wireless headphones may be ultra-portable playback devices that are powered by an internal energy source and weigh less than fifty ounces.

(61) It should be appreciated that the playback device **102** may take the form of other wearable devices separate and apart from a headphone. Wearable devices may include those devices configured to be worn about a portion of a subject (e.g., a head, a neck, a torso, an arm, a wrist, a finger, a leg, an ankle, etc.). For example, the playback device **102** may take the form of a pair of glasses including a frame front (e.g., configured to hold one or more lenses), a first temple rotatably coupled to the frame front, and a second temple rotatable coupled to the frame front. In this example, the pair of glasses may comprise one or more transducers integrated into at least one of the first and second temples and configured to project sound towards an ear of the subject.

(62) While specific implementations of playback and network microphone devices have been described above, including implementations described with respect to FIGS. 2A and 2B, there are numerous configurations of devices, including, but not limited to, those having no UI, microphones in different locations, multiple microphone arrays positioned in different arrangements, and/or any other configuration as appropriate to the requirements of a given application. For example, UIs and/or microphone arrays can be implemented in other playback devices and/or computing devices rather than those described herein. Further, although a specific example of playback device **102** is described with reference to MPS **100**, one skilled in the art will recognize that playback devices as described herein can be used in a variety of different environments, including (but not limited to) environments with more and/or fewer elements, without departing from this invention. Likewise, MPS's as described herein can be used with various different playback devices.

(63) By way of illustration, SONOS, Inc. presently offers (or has offered) for sale certain playback devices that may implement certain of the embodiments disclosed herein, including a "SONOS ONE," "PLAY:1," "PLAY:3," "PLAY:5," "PLAYBAR," "AMP," "CONNECT:AMP," "PLAYBASE," "BEAM," "CONNECT," "SUB," and "SONOS MOVE." Any other past, present, and/or future playback devices may additionally or alternatively be used to implement the playback devices of example embodiments disclosed herein. Additionally, it should be understood that a playback device is not limited to the examples illustrated in FIG. 2A or 2B, or to the SONOS product offerings. For example, a playback device may be integral to another device or component such as a television, a lighting fixture, or some other device for indoor or outdoor use.

(64) b. Example Playback Device Configurations

(65) FIGS. 3A-3E show example configurations of playback devices. Referring first to FIG. 3A, in some example instances, a single playback device may belong to a zone. For example, the playback device **102c** (FIG. 1A) on the Patio may belong to Zone A. In some implementations described below, multiple playback devices may be “bonded” to form a “bonded pair,” which together form a single zone. For example, the playback device **102f** (FIG. 1A) named “Bed 1” in FIG. 3A may be bonded to the playback device **102g** (FIG. 1A) named “Bed 2” in FIG. 3A to form Zone B. Bonded playback devices may have different playback responsibilities (e.g., channel responsibilities). In another implementation described below, multiple playback devices may be merged to form a single zone. For example, the playback device **102d** named “Bookcase” may be merged with the playback device **102m** named “Living Room” to form a single Zone C. The merged playback devices **102d** and **102m** may not be specifically assigned different playback responsibilities. That is, the merged playback devices **102d** and **102m** may, aside from playing audio content in synchrony, each play audio content as they would if they were not merged.

(66) For purposes of control, each zone in the MPS **100** may be represented as a single user interface (“UI”) entity. For example, as displayed by the controller devices **104**, Zone A may be provided as a single entity named “Portable,” Zone B may be provided as a single entity named “Stereo,” and Zone C may be provided as a single entity named “Living Room.”

(67) In various embodiments, a zone may take on the name of one of the playback devices belonging to the zone. For example, Zone C may take on the name of the Living Room device **102m** (as shown). In another example, Zone C may instead take on the name of the Bookcase device **102d**. In a further example, Zone C may take on a name that is some combination of the Bookcase device **102d** and Living Room device **102m**. The name that is chosen may be selected by a user via inputs at a controller device **104**. In some embodiments, a zone may be given a name that is different than the device(s) belonging to the zone. For example, Zone B in FIG. 3A is named “Stereo” but none of the devices in Zone B have this name. In one aspect, Zone B is a single UI entity representing a single device named “Stereo,” composed of constituent devices “Bed 1” and “Bed 2.” In one implementation, the Bed 1 device may be playback device **102f** in the master bedroom **101h** (FIG. 1A) and the Bed 2 device may be the playback device **102g** also in the master bedroom **101h** (FIG. 1A).

(68) As noted above, playback devices that are bonded may have different playback responsibilities, such as playback responsibilities for certain audio channels. For example, as shown in FIG. 3B, the Bed 1 and Bed 2 devices **102f** and **102g** may be bonded so as to produce or enhance a stereo effect of audio content. In this example, the Bed 1 playback device **102f** may be configured to play a left channel audio component, while the Bed 2 playback device **102g** may be configured to play a right channel audio component. In some implementations, such stereo bonding may be referred to as “pairing.”

(69) Additionally, playback devices that are configured to be bonded may have additional and/or different respective speaker drivers. As shown in FIG. 3C, the playback device **102b** named “Front” may be bonded with the playback device **102k** named “SUB.” The Front device **102b** may render a range of mid to high frequencies, and the SUB device **102k** may render low frequencies as, for example, a subwoofer. When unbonded, the Front device **102b** may be configured to render a full range of frequencies. As another example, FIG. 3D shows the Front and SUB devices **102b** and **102k** further bonded with Right and Left playback devices **102a** and **102j**, respectively. In some implementations, the Right and Left devices **102a** and **102j** may form surround or “satellite” channels of a home theater system. The bonded playback devices **102a**, **102b**, **102j**, and **102k** may form a single Zone D (FIG. 3A).

(70) In some implementations, playback devices may also be “merged.” In contrast to certain bonded playback devices, playback devices that are merged may not have assigned playback responsibilities, but may each render the full range of audio content that each respective playback

device is capable of. Nevertheless, merged devices may be represented as a single UI entity (i.e., a zone, as discussed above). For instance, FIG. 3E shows the playback devices **102d** and **102m** in the Living Room merged, which would result in these devices being represented by the single UI entity of Zone C. In one embodiment, the playback devices **102d** and **102m** may playback audio in synchrony, during which each outputs the full range of audio content that each respective playback device **102d** and **102m** is capable of rendering.

(71) In some embodiments, a stand-alone NMD may be in a zone by itself. For example, the NMD **103h** from FIG. 1A is named “Closet” and forms Zone I in FIG. 3A. An NMD may also be bonded or merged with another device so as to form a zone. For example, the NMD device **103f** named “Island” may be bonded with the playback device **102i** Kitchen, which together form Zone F, which is also named “Kitchen.” Additional details regarding assigning NMDs and playback devices as designated or default devices may be found, for example, in previously referenced U.S. Patent Publication No. 2017-0242653. In some embodiments, a stand-alone NMD may not be assigned to a zone.

(72) Zones of individual, bonded, and/or merged devices may be arranged to form a set of playback devices that playback audio in synchrony. Such a set of playback devices may be referred to as a “group,” “zone group,” “synchrony group,” or “playback group.” In response to inputs provided via a controller device **104**, playback devices may be dynamically grouped and ungrouped to form new or different groups that synchronously play back audio content. For example, referring to FIG. 3A, Zone A may be grouped with Zone B to form a zone group that includes the playback devices of the two zones. As another example, Zone A may be grouped with one or more other Zones C-I. The Zones A-I may be grouped and ungrouped in numerous ways. For example, three, four, five, or more (e.g., all) of the Zones A-I may be grouped. When grouped, the zones of individual and/or bonded playback devices may play back audio in synchrony with one another, as described in previously referenced U.S. Pat. No. 8,234,395. Grouped and bonded devices are example types of associations between portable and stationary playback devices that may be caused in response to a trigger event, as discussed above and described in greater detail below.

(73) In various implementations, the zones in an environment may be assigned a particular name, which may be the default name of a zone within a zone group or a combination of the names of the zones within a zone group, such as “Dining Room+Kitchen,” as shown in FIG. 3A. In some embodiments, a zone group may be given a unique name selected by a user, such as “Nick's Room,” as also shown in FIG. 3A. The name “Nick's Room” may be a name chosen by a user over a prior name for the zone group, such as the room name “Master Bedroom.”

(74) Referring back to FIG. 2A, certain data may be stored in the memory **213** as one or more state variables that are periodically updated and used to describe the state of a playback zone, the playback device(s), and/or a zone group associated therewith. The memory **213** may also include the data associated with the state of the other devices of the media playback system **100**, which may be shared from time to time among the devices so that one or more of the devices have the most recent data associated with the system.

(75) In some embodiments, the memory **213** of the playback device **102** may store instances of various variable types associated with the states. Variables instances may be stored with identifiers (e.g., tags) corresponding to type. For example, certain identifiers may be a first type “a1” to identify playback device(s) of a zone, a second type “b1” to identify playback device(s) that may be bonded in the zone, and a third type “c1” to identify a zone group to which the zone may belong. As a related example, in FIG. 1A, identifiers associated with the Patio may indicate that the Patio is the only playback device of a particular zone and not in a zone group. Identifiers associated with the Living Room may indicate that the Living Room is not grouped with other zones but includes bonded playback devices **102a**, **102b**, **102j**, and **102k**. Identifiers associated with the Dining Room may indicate that the Dining Room is part of Dining Room+Kitchen group and that devices **103f** and **102i** are bonded. Identifiers associated with the Kitchen may indicate the same or similar

information by virtue of the Kitchen being part of the Dining Room+Kitchen zone group. Other example zone variables and identifiers are described below.

(76) In yet another example, the MPS **100** may include variables or identifiers representing other associations of zones and zone groups, such as identifiers associated with Areas, as shown in FIG. 3A. An Area may involve a cluster of zone groups and/or zones not within a zone group. For instance, FIG. 3A shows a first area named “First Area” and a second area named “Second Area.” The First Area includes zones and zone groups of the Patio, Den, Dining Room, Kitchen, and Bathroom. The Second Area includes zones and zone groups of the Bathroom, Nick's Room, Bedroom, and Living Room. In one aspect, an Area may be used to invoke a cluster of zone groups and/or zones that share one or more zones and/or zone groups of another cluster. In this respect, such an Area differs from a zone group, which does not share a zone with another zone group. Further examples of techniques for implementing Areas may be found, for example, in U.S. Patent Publication No. 2018-0107446 published Apr. 19, 2018 and titled “Room Association Based on Name,” and U.S. Pat. No. 8,483,853 filed Sep. 11, 2007, and titled “Controlling and manipulating groupings in a multi-zone media system,” each of which is incorporated herein by reference in its entirety. In some embodiments, the MPS **100** may not implement Areas, in which case the system may not store variables associated with Areas.

(77) The memory **213** may be further configured to store other data. Such data may pertain to audio sources accessible by the playback device **102** or a playback queue that the playback device (or some other playback device(s)) may be associated with. In embodiments described below, the memory **213** is configured to store a set of command data for selecting a particular VAS when processing voice inputs.

(78) During operation, one or more playback zones in the environment of FIG. 1A may each be playing different audio content. For instance, the user may be grilling in the Patio zone and listening to hip hop music being played by the playback device **102c**, while another user may be preparing food in the Kitchen zone and listening to classical music being played by the playback device **102i**. In another example, a playback zone may play the same audio content in synchrony with another playback zone. For instance, the user may be in the Office zone where the playback device **102n** is playing the same hip-hop music that is being playing by playback device **102c** in the Patio zone. In such a case, playback devices **102c** and **102n** may be playing the hip-hop in synchrony such that the user may seamlessly (or at least substantially seamlessly) enjoy the audio content that is being played out-loud while moving between different playback zones. Synchronization among playback zones may be achieved in a manner similar to that of synchronization among playback devices, as described in previously referenced U.S. Pat. No. 8,234,395.

(79) As suggested above, the zone configurations of the MPS **100** may be dynamically modified. As such, the MPS **100** may support numerous configurations. For example, if a user physically moves one or more playback devices to or from a zone, the MPS **100** may be reconfigured to accommodate the change(s). For instance, if the user physically moves the playback device **102c** from the Patio zone to the Office zone, the Office zone may now include both the playback devices **102c** and **102n**. In some cases, the user may pair or group the moved playback device **102c** with the Office zone and/or rename the players in the Office zone using, for example, one of the controller devices **104** and/or voice input. As another example, if one or more playback devices **102** are moved to a particular space in the home environment that is not already a playback zone, the moved playback device(s) may be renamed or associated with a playback zone for the particular space.

(80) Further, different playback zones of the MPS **100** may be dynamically combined into zone groups or split up into individual playback zones. For example, the Dining Room zone and the Kitchen zone may be combined into a zone group for a dinner party such that playback devices **102i** and **102l** may render audio content in synchrony. As another example, bonded playback

devices in the Den zone may be split into (i) a television zone and (ii) a separate listening zone. The television zone may include the Front playback device **102b**. The listening zone may include the Right, Left, and SUB playback devices **102a**, **102j**, and **102k**, which may be grouped, paired, or merged, as described above. Splitting the Den zone in such a manner may allow one user to listen to music in the listening zone in one area of the living room space, and another user to watch the television in another area of the living room space. In a related example, a user may utilize either of the NMD **103a** or **103b** (FIG. **1B**) to control the Den zone before it is separated into the television zone and the listening zone. Once separated, the listening zone may be controlled, for example, by a user in the vicinity of the NMD **103a**, and the television zone may be controlled, for example, by a user in the vicinity of the NMD **103b**. As described above, however, any of the NMDs **103** may be configured to control the various playback and other devices of the MPS **100**.

(81) c. Example Controller Devices

(82) FIG. **4A** is a functional block diagram illustrating certain aspects of a selected one of the controller devices **104** of the MPS **100** of FIG. **1A**. Controller devices in accordance with several embodiments of the invention can be used in various systems, such as (but not limited to) an MPS as described in FIG. **1A**. Such controller devices may also be referred to herein as a “control device” or “controller.” The controller device shown in FIG. **4A** may include components that are generally similar to certain components of the network devices described above, such as a processor **412**, memory **413** storing program software **414**, at least one network interface **424**, and one or more microphones **422**. In one example, a controller device may be a dedicated controller for the MPS **100**. In another example, a controller device may be a network device on which media playback system controller application software may be installed, such as for example, an iPhone™, iPad™ or any other smart phone, tablet, or network device (e.g., a networked computer such as a PC or Mac™).

(83) The memory **413** of the controller device **104** may be configured to store controller application software and other data associated with the MPS **100** and/or a user of the system **100**. The memory **413** may be loaded with instructions in software **414** that are executable by the processor **412** to achieve certain functions, such as facilitating user access, control, and/or configuration of the MPS **100**. The controller device **104** may be configured to communicate with other network devices via the network interface **424**, which may take the form of a wireless interface, as described above.

(84) In one example, system information (e.g., such as a state variable) may be communicated between the controller device **104** and other devices via the network interface **424**. For instance, the controller device **104** may receive playback zone and zone group configurations in the MPS **100** from a playback device, an NMD, or another network device. Likewise, the controller device **104** may transmit such system information to a playback device or another network device via the network interface **424**. In some cases, the other network device may be another controller device.

(85) The controller device **104** may also communicate playback device control commands, such as volume control and audio playback control, to a playback device via the network interface **424**. As suggested above, changes to configurations of the MPS **100** may also be performed by a user using the controller device **104**. The configuration changes may include adding/removing one or more playback devices to/from a zone, adding/removing one or more zones to/from a zone group, forming a bonded or merged player, separating one or more playback devices from a bonded or merged player, among others.

(86) As shown in FIG. **4A**, the controller device **104** may also include a user interface **440** that is generally configured to facilitate user access and control of the MPS **100**. The user interface **440** may include a touch-screen display or other physical interface configured to provide various graphical controller interfaces, such as the controller interfaces **440a** and **440b** shown in FIGS. **4B** and **4C**. Referring to FIGS. **4B** and **4C** together, the controller interfaces **440a** and **440b** include a playback control region **442**, a playback zone region **443**, a playback status region **444**, a playback queue region **446**, and a sources region **448**. The user interface as shown is just one example of an

interface that may be provided on a network device, such as the controller device shown in FIG. 4A, and accessed by users to control a media playback system, such as the MPS **100**. Other user interfaces of varying formats, styles, and interactive sequences may alternatively be implemented on one or more network devices to provide comparable control access to a media playback system. (87) The playback control region **442** (FIG. 4B) may include selectable icons (e.g., by way of touch or by using a cursor) that, when selected, cause playback devices in a selected playback zone or zone group to play or pause, fast forward, rewind, skip to next, skip to previous, enter/exit shuffle mode, enter/exit repeat mode, enter/exit cross fade mode, etc. The playback control region **442** may also include selectable icons that, when selected, modify equalization settings and/or playback volume, among other possibilities.

(88) The playback zone region **443** (FIG. 4C) may include representations of playback zones within the MPS **100**. The playback zones regions **443** may also include a representation of zone groups, such as the Dining Room+Kitchen zone group, as shown. In some embodiments, the graphical representations of playback zones may be selectable to bring up additional selectable icons to manage or configure the playback zones in the MPS **100**, such as a creation of bonded zones, creation of zone groups, separation of zone groups, and renaming of zone groups, among other possibilities.

(89) For example, as shown, a “group” icon may be provided within each of the graphical representations of playback zones. The “group” icon provided within a graphical representation of a particular zone may be selectable to bring up options to select one or more other zones in the MPS **100** to be grouped with the particular zone. Once grouped, playback devices in the zones that have been grouped with the particular zone will be configured to play audio content in synchrony with the playback device(s) in the particular zone. Analogously, a “group” icon may be provided within a graphical representation of a zone group. In this case, the “group” icon may be selectable to bring up options to deselect one or more zones in the zone group to be removed from the zone group. Other interactions and implementations for grouping and ungrouping zones via a user interface are also possible. The representations of playback zones in the playback zone region **443** (FIG. 4C) may be dynamically updated as playback zone or zone group configurations are modified.

(90) The playback status region **444** (FIG. 4B) may include graphical representations of audio content that is presently being played, previously played, or scheduled to play next in the selected playback zone or zone group. The selected playback zone or zone group may be visually distinguished on a controller interface, such as within the playback zone region **443** and/or the playback status region **444**. The graphical representations may include track title, artist name, album name, album year, track length, and/or other relevant information that may be useful for the user to know when controlling the MPS **100** via a controller interface.

(91) The playback queue region **446** may include graphical representations of audio content in a playback queue associated with the selected playback zone or zone group. In some embodiments, each playback zone or zone group may be associated with a playback queue comprising information corresponding to zero or more audio items for playback by the playback zone or zone group. For instance, each audio item in the playback queue may comprise a uniform resource identifier (URI), a uniform resource locator (URL), or some other identifier that may be used by a playback device in the playback zone or zone group to find and/or retrieve the audio item from a local audio content source or a networked audio content source, which may then be played back by the playback device.

(92) In one example, a playlist may be added to a playback queue, in which case information corresponding to each audio item in the playlist may be added to the playback queue. In another example, audio items in a playback queue may be saved as a playlist. In a further example, a playback queue may be empty, or populated but “not in use” when the playback zone or zone group is playing continuously streamed audio content, such as Internet radio that may continue to play

until otherwise stopped, rather than discrete audio items that have playback durations. In an alternative embodiment, a playback queue can include Internet radio and/or other streaming audio content items and be “in use” when the playback zone or zone group is playing those items. Other examples are also possible.

(93) When playback zones or zone groups are “grouped” or “ungrouped,” playback queues associated with the affected playback zones or zone groups may be cleared or re-associated. For example, if a first playback zone including a first playback queue is grouped with a second playback zone including a second playback queue, the established zone group may have an associated playback queue that is initially empty, that contains audio items from the first playback queue (such as if the second playback zone was added to the first playback zone), that contains audio items from the second playback queue (such as if the first playback zone was added to the second playback zone), or a combination of audio items from both the first and second playback queues. Subsequently, if the established zone group is ungrouped, the resulting first playback zone may be re-associated with the previous first playback queue or may be associated with a new playback queue that is empty or contains audio items from the playback queue associated with the established zone group before the established zone group was ungrouped. Similarly, the resulting second playback zone may be re-associated with the previous second playback queue or may be associated with a new playback queue that is empty or contains audio items from the playback queue associated with the established zone group before the established zone group was ungrouped. Other examples are also possible.

(94) With reference still to FIGS. 4B and 4C, the graphical representations of audio content in the playback queue region **446** (FIG. 4B) may include track titles, artist names, track lengths, and/or other relevant information associated with the audio content in the playback queue. In one example, graphical representations of audio content may be selectable to bring up additional selectable icons to manage and/or manipulate the playback queue and/or audio content represented in the playback queue. For instance, a represented audio content may be removed from the playback queue, moved to a different position within the playback queue, or selected to be played immediately, or after any currently playing audio content, among other possibilities. A playback queue associated with a playback zone or zone group may be stored in a memory on one or more playback devices in the playback zone or zone group, on a playback device that is not in the playback zone or zone group, and/or some other designated device. Playback of such a playback queue may involve one or more playback devices playing back media items of the queue, perhaps in sequential or random order.

(95) The sources region **448** may include graphical representations of selectable audio content sources and/or selectable voice assistants associated with a corresponding VAS. The VASes may be selectively assigned. In some examples, multiple VASes, such as AMAZON's Alexa, MICROSOFT's Cortana, etc., may be invocable by the same NMD. In some embodiments, a user may assign a VAS exclusively to one or more NMDs. For example, a user may assign a first VAS to one or both of the NMDs **102a** and **102b** in the Living Room shown in FIG. 1A, and a second VAS to the NMD **103f** in the Kitchen. Other examples are possible.

(96) d. Example Audio Content Sources

(97) The audio sources in the sources region **448** may be audio content sources from which audio content may be retrieved and played by the selected playback zone or zone group. One or more playback devices in a zone or zone group may be configured to retrieve for playback audio content (e.g., according to a corresponding URI or URL for the audio content) from a variety of available audio content sources. In one example, audio content may be retrieved by a playback device directly from a corresponding audio content source (e.g., via a line-in connection). In another example, audio content may be provided to a playback device over a network via one or more other playback devices or network devices. As described in greater detail below, in some embodiments, audio content may be provided by one or more media content services.

(98) Example audio content sources may include a memory of one or more playback devices in a

media playback system such as the MPS **100** of FIG. **1**, local music libraries on one or more network devices (e.g., a controller device, a network-enabled personal computer, or a networked-attached storage (“NAS”)), streaming audio services providing audio content via the Internet (e.g., cloud-based music services), or audio sources connected to the media playback system via a line-in input connection on a playback device or network device, among other possibilities.

(99) In some embodiments, audio content sources may be added or removed from a media playback system such as the MPS **100** of FIG. **1A**. In one example, an indexing of audio items may be performed whenever one or more audio content sources are added, removed, or updated.

Indexing of audio items may involve scanning for identifiable audio items in all folders/directories shared over a network accessible by playback devices in the media playback system and generating or updating an audio content database comprising metadata (e.g., title, artist, album, track length, among others) and other associated information, such as a URI or URL for each identifiable audio item found. Other examples for managing and maintaining audio content sources may also be possible.

III. Example Simulation of Network Device Removal

(100) As previously discussed, a media playback system may include one or more network devices (e.g., playback devices, networking components, smart devices, etc.) that may no longer be necessary for the operation of the media playback system due to the addition of newer network devices to the media playback system, or potentially a software and/or firmware update to the media playback system that renders one or more of the network devices unusable. However, the user may not know which network devices can be removed from the media playback system while still allowing the media playback system to operate as desired. As one solution, a user may attempt to disconnect one or more of the network devices and then assess whether the media playback system is still operational. However, this approach may be tedious and might not exactly identify which of the network devices can be removed from the media playback system without crippling its operation. This may be due in part to different network topologies that may be used to connect each network device to a particular media playback system.

(101) For instance, as one example, a media playback system may have a network topology in which one or more of the network devices (e.g., bridging device or hub) facilitates connection of the network devices in the media playback system. In this example, the media playback system may be configured to have a particular network device connected via ethernet to a wireless router and/or access point and then, in turn, the particular network device may be configured to serve as an intermediate connection point by creating a wireless mesh network to which other network devices in the media playback system can connect. In this respect, if a user disconnects the particular network device from the media playback system, the other network devices may cease to function as desired. However, short of manually disconnecting each network device and determining whether the media playback system remains functional for the remaining connected devices, the user may not be able to confidently remove devices from the media playback system.

(102) To provide the user with more confidence when removing devices from the media playback system, a user device may simulate the removal of one or more network devices from the media playback system and then provide the user with an indication of whether the remaining network devices remain connected to the media playback system, thereby determining whether one or more network devices can be removed while still allowing the media playback system to operate normally.

(103) FIG. **5** depicts a message flow diagram **500** between a user device **502**, a first set of network devices **504**, and a second set of network devices **506** for simulating the removal of the first set of network devices **504**. The user device **502** may include one or more of controller devices **104a-b** as described with respect to FIG. **1A**, and/or controller device **104** as described with respect to FIGS. **4A-4C** above. The first set of network devices **504** may be one or more network devices that will be evaluated for disconnection from a media playback system, which may include any of the

playback devices **102a-102o** described above with respect to FIGS. **1A-3B** above. The first set of network devices **504** may also include one or more non-playback network devices as well, including but not limited to a bridging device, such as the “SONOS BRIDGE,” and/or a Wi-Fi extender device, such as the “SONOS BOOST.”

(104) The second set of network devices **506** shown in FIG. **5** may be one or more network devices that a user desires to keep connected to the media playback system, which may also include any of the playback devices **102a-102o** described above with respect to FIGS. **1A-3B** above. The user device **502**, the first set of network devices **504**, and the second set of network devices **506** may be communicatively coupled based on a network topology of the media playback system. The network topology may take various forms, some examples of which are described further below with respect to FIGS. **6A-6D**.

(105) Further, it should be understood that while this disclosure discusses examples with respect to network devices within media playback systems, this is for purposes of illustration only and the functions described herein may also be applied to any networked system of devices (e.g., “smart” home networks that include Internet of Things (IoT) devices such as smart thermostats, smart light bulbs, smart outlets, smart appliances, smart fixtures, etc.).

(106) The user device **502**, the first set of network devices **504**, and the second set of network devices **506** for simulating the removal of the first set of network devices **504** may be connected to one another as part of a network topology of the media playback system. However, depending on the configuration of the network topology, the user device **502** may perform pre-simulation steps before the user device **502** begins performing the functions in the message flow diagram **500**. For instance, as one example, if the first set of network devices **504** includes a bridge device that creates a mesh network to which the second set of network devices **506** connect, the second set of network devices **506** may require network credentials in order to perform the simulation described with respect to the message flow diagram **500**.

(107) To provide these network credentials, the user device **502** may first request network credentials from a user. The network credentials may include Wi-Fi credentials that permit connection to an accessible LAN (e.g., Wi-Fi network). The network credentials may take various forms. As one possibility, the network credentials may include (i) a service set identifier (SSID) and (ii) a password. In this respect, the SSID may include the name of a Wi-Fi network and the password may include a string of characters that complies with one of the wired equivalent privacy (WEP), Wi-Fi protected access (WPA), or Wi-Fi protected access version 2 (WPA2) security methods. The network credentials may take other forms as well.

(108) The user device **502** may then test the network credentials, which may take various forms. As one possibility, testing the network credentials may involve the user device **502** transmitting the network credentials to one or more network devices of the first set of network devices **504** along with a command configured to cause the one or more network devices of the first set of network devices **504** to attempt to establish a connection to the Wi-Fi network.

(109) However, in some circumstances, the first set of network devices may not have the capability to connect to the Wi-Fi network because of hardware or software limitations. Thus, as another possibility, testing the network credentials may involve the user device **502** transmitting the network credentials to one or more network devices of the second set of network devices **506** along with a command configured to cause the one or more network devices of the second set of network devices **506** to attempt to establish a connection to the Wi-Fi network. In this respect, if the one or more network devices of the second set of network devices **506** is able to establish a connection to the Wi-Fi network, the user device **502** can infer that the network credentials are valid.

(110) As yet another possibility, testing the network credentials may involve the user device **502** establishing a connection with the Wi-Fi network itself. In this respect, the user device **502** can validate the network credentials without the need of transmitting the credentials to the first set of network devices **504** or the second set of network devices **506** for testing.

(111) The function of testing the network credentials may take other forms as well.

(112) The user device **502** may then transmit the network credentials to both the first set of network devices **504** and the second set of network devices **506**. The function of transmitting the network credentials serves to provide the first set of network devices **504** and the second set of network devices **506** with the ability to connect to the media playback system directly through the Wi-Fi network, which may be necessary depending on the topology of the media playback system.

(113) Once the user device **502** transmits the network credentials to each network device in the first set of network devices **504** and the second set of network devices **506**, the user device **502** can begin performing the steps associated with the message flow diagram **500**.

(114) At block **508**, the user device **502** may transmit, at an initialization time, a command to the first set of network devices **504** to, at a first time, temporarily disconnect from the other network devices in the media playback system via a particular communication protocol.

(115) The command the user device **502** transmits to the first set of network devices **504** may take various forms. As one possibility, the command may include a command to disconnect from the other network devices in the media playback system via a particular communication protocol after a first delay. In this respect, the particular communication protocol may take various forms. As one example, the particular communication protocol may include a wired communication protocol. As another example, the particular communication protocol may include a wireless communication protocol. The particular communication protocol may take other forms as well.

(116) Further, the particular communication protocol via which the first set of network devices **504** should disconnect may depend on the network topology of the media playback system. For instance, if the first set of network devices **504** are connected to the other network devices in the media playback system via a wired communication protocol, then the command to disconnect from the other network devices in the media playback system via a particular communication protocol may include a command to disconnect from the media playback system via the wired communication protocol. Likewise, if the first set of network devices **504** are connected to the other network devices in the media playback system via a wireless communication protocol, then the command to disconnect from the media playback system via a particular communication protocol may include a command to disconnect from the other network devices in the media playback system via the wireless communication protocol.

(117) As another possibility, the command the user device **502** transmits to the first set of network devices **504** may include a command to later reconnect to the other network devices in the media playback system via the particular communication protocol. In this respect, the period of time that the first set of network devices **504** is disconnected may be long enough (e.g., one to two minutes) to allow the second set of network devices **506** to test network connectivity to the other network devices in the media playback system, which is discussed in detail below with respect to block **510**. This may be beneficial in simulating the removal of the first set of network devices **504** insofar as the first set of network devices **504** will remain disconnected long enough for the second set of network devices **506** to complete their network connectivity test.

(118) The command the user device **502** transmits to the first set of network devices **504** may take other forms as well.

(119) As shown at block **510**, the user device **502** may transmit a command to each network device of the second set of network devices **506** to, at a second time, test their network connectivity with other network devices in the media playback system. The function of testing their network connectivity may take various forms. As one possibility, testing network connectivity may involve each network device of the second set of network devices **506** querying the user device **502** via the LAN. In this respect, if each network device of the second set of network devices **506** is able to query the user device **502**, then each network device of the second set of network devices **506** may determine that the network connectivity test was successful. As another possibility, testing network connectivity may involve each network device of the second set of network devices **506**

establishing connection to a website (e.g., on the Internet) via the LAN. In this respect, each network device of the second set of network devices **506** may determine that the network connectivity test was successful if it is able to receive one or more data packets from the website. The function of each network device of the second set of network devices **506** testing its network connectivity may take other forms as well.

(120) In some circumstances, one or more of the network devices in the second set of network devices **506** may determine that the network connectivity test was unsuccessful. These circumstances may depend on the network topology of the media playback system via which the second set of network devices **506** is connected. As one example, one or more of the first set of network devices **504** may have served as a gateway, access point, or bridge to the media playback system for one or more network devices in the second set of network devices **506**. As another example, one network device of the first set of network devices **504** may have created a wireless mesh network to which one or more of the network devices of the second set of network devices **506** is connected. As yet another example, one or more network devices of the first set of network devices **504** may have served as one or more nodes of a wireless mesh network through which one of the network devices of the second set of network devices **506** achieved its connection to the other network devices in the media playback system. In this respect, the first set of network devices **504** carrying out the command to disconnect from the other network devices in the media playback system as discussed above with respect to block **510** may cause the network connectivity test of one or more network devices in the second set of network devices **506** to be unsuccessful.

(121) Further, similar to the command discussed above with respect to block **514**, the user device **502** may command each network device of the second set of network devices **506** to test its network connectivity with other network devices in the media playback system after a second delay. The second delay may be longer than the first delay, and long enough to allow each network device of the first set of network devices **504** to fully disconnect from the other network devices in the media playback system via the particular communication protocol. In this respect, the second delay increases the likelihood that each network device in the second set of network devices **506** tests its network connectivity during the simulated removal of the first set of network devices **504** from the media playback system.

(122) As shown at block **512**, the first set of network devices **504** may, at the first time, temporarily disconnect from the LAN. As previously discussed, the function of disconnecting from the LAN may involve temporarily disconnecting from the LAN via a particular communication protocol (e.g., a wired or wireless communication protocol) depending on the network topology of the media playback system. Further, the first set of network devices **504** temporarily disconnecting from the LAN may allow the second set of network devices **506** to properly test their network connectivity under the desired conditions.

(123) As shown at block **514**, the second set of network devices **506** may, at the second time, test their network connectivity to the other network devices in the media playback system. As previously discussed with respect to block **510**, the function of testing their network connectivity may include, after the second delay, (i) querying the user device **102**, (ii) establishing a connection to a website (e.g., on the Internet), (iii) or an equivalent method.

(124) As shown at block **516**, the first set of network devices **504** may, at a third time, reconnect to the other network devices in the media playback system. As previously discussed with respect to block **508**, the function of reconnecting to the other network devices in the media playback system may include reconnecting to the LAN via the particular communication protocol (e.g., wired or wireless communication protocol) after a third delay, which is longer than the second delay and long enough to allow the network connectivity test performed at block **514** to be completed.

(125) As shown at block **518**, the user device **502** may request network connectivity results from each network device in the second set of network device **506**. The network connectivity results may take various forms. As one possibility, the network connectivity results may take the form of a

success or failure indication. In this respect, the network connectivity results may simply indicate whether each network device in the second set of network devices **506** was able to establish a connection with other network devices in the media playback device and, in turn, the user device **502**. As another possibility, the network connectivity results may take the form of a detailed network analysis that includes information including, but not limited to (i) network connection strength, (ii) bandwidth, and (iii) latency. The network connectivity results may take other forms as well.

(126) As shown at block **520**, each network device in the second set of network devices **506** may transmit its network connectivity test results to the user device **502**. Upon receiving the network connectivity results, the user device **502** may determine whether the simulation of removing the first set of network devices **504** was successful or unsuccessful. This determination may take various forms. As one possibility, if all of the network connectivity test results indicate a successful connection with other network devices in the media playback system, then the user device **502** may determine that the simulation of removing the first set of network devices **504** was successful. As another possibility, if just one of the network connectivity test results indicates that a network device was unable to connect to other network devices in the media playback system while the first set of network devices **504** were disconnected, then the user device **502** may determine that the simulation of removing the first set of network devices **504** was unsuccessful. The determination of whether the simulation of removing the first set of network devices **504** was successful or unsuccessful may take other forms as well.

(127) Once the user device **502** has determined whether the simulation of removing the first set of network devices **504** was successful or unsuccessful, at block **522**, the user device **502** may display the network connectivity results to the user. The network connectivity results that are displayed to the user may take various forms that may depend on whether the user device **502** determined that the simulation of removing the first set of network devices **504** was successful or unsuccessful. As one possibility, if the user device **502** determined that the simulation was successful, the network connectivity results that are displayed to the user may take the form of an indication that informs the user that the user can confidently remove the first set of network devices **504** from the media playback system without affecting performance of the media playback system. As another possibility, if the user device **502** determined that the simulation was unsuccessful, the network connectivity results that are displayed to the user may take the form of an indication that informs the user that the user cannot remove the first set of network devices **504** from the media playback system without affecting performance of the media playback system. As yet another possibility, if the user device **502** determined that the simulation was unsuccessful, the network connectivity results that are displayed to the user may take the form of troubleshooting instructions that inform the user of potential adjustments that can be made to the media playback system in an attempt to improve the network connectivity results. As still yet another possibility, the user device **502** may determine that the simulation of removing a first subset of network devices in the first set of network devices **504** was unsuccessful, but that the simulation of removing a second subset of network devices of the first set of network devices **504** was successful. In this respect, the network connectivity results that are displayed to the user may take the form of an indication that informs the user that the user (i) can remove the second subset of network devices of the first set of network devices **504** without affecting performance of the media playback system, and (ii) cannot remove the first subset of network devices of the first set of network devices **504** without affecting performance of the media playback system. The network connectivity results that are displayed to the user may take other forms as well.

(128) As mentioned above, some aspects of the method flow diagram **500** may change or be affected by a given media playback system's network topology. As such, FIGS. **6A-6D** provide illustrative examples of different media playback system network topologies and how the method flow diagram **500** may be applied to each.

(129) FIG. 6A depicts a media playback system network topology **600** that includes a user device **602**, a router **604**, and network devices **606**, **608**, **610**, and **612**. The user device **602** may include one or more of controller devices **104** as described with respect to FIG. 1A, controller device **104** as described with respect to FIGS. 4A-4C above, as well as the user device **502** as described with respect to FIG. 5. Further, as shown in FIG. 6A, the network device **606** may be a wireless bridge that is configured to connect the network devices **608**, **610**, and **612** to the LAN via a wireless mesh network created by the network device **606**. The network devices **608**, **610**, and **612** may include any of playback devices **102a-102o** as described with respect to FIGS. 1A-3E, but notably, the network device **608** may lack an ability to connect to the LAN without the mesh network created by the network device **606**, while network devices **610** and **612** may have the ability to connect to the LAN independently of the network device **606**.

(130) As shown, the network topology **600** includes the user device **602** wirelessly connected to the router **604** via a wireless communication protocol (denoted by a dashed line) and the network device **606** connected to the router **604** via a wired communication protocol (denoted by a solid line).

(131) In practice, applying the steps of the method flow diagram **500** described above with respect to FIG. 5 to the network topology **600** may first include the user device **602**, after validating and transmitting the network credentials of the LAN, transmitting a command to the network device **606** to temporarily disconnect from the other network devices (e.g., the user device **602** and the network devices **608**, **610**, and **612**) in the media playback system via the wired communication protocol. As previously discussed, this command might be sent at an initialization time and may include a command for the network device **606** to, at a first time, temporarily disable a network interface that corresponds to the wired communication protocol (e.g., temporarily switch off its ethernet port). The command might also include a command to, after a delay, reenabling the network interface that corresponds to the wired communication protocol.

(132) The user device **602** may also transmit a command to each of the network devices **608**, **610**, and **612** to, at a second time, test its network connectivity with other network devices in the media playback system. In this example, the second time is after the first time at which the network device **606** temporarily disconnects from the other network devices in the media playback system but before a third time at which the network device **606** reconnects to the other network devices in the media playback system.

(133) The network device **606** may then, at the first time, temporarily disconnect from the other network devices in the media playback system via the LAN, thereby disconnecting the wireless mesh network.

(134) Then, at the second time, each of the network devices **608**, **610**, and **612** may perform a network connectivity test, which may take the form of attempting to query the user device **602** via the LAN. As previously mentioned, network device **608** may be unable to query the user device **602** due to an inability to connect to the LAN while the network device **606** is temporarily disconnected. Meanwhile, network devices **610** and **612** may query the user device **602** without issue as the network devices **610** and **612** have the capability to connect to the LAN independent of the wireless mesh network created by the network device **606**.

(135) Next, at the third time, the network device **606** may reconnect to the LAN and thereby reestablish the wireless mesh network. In turn, the network device **608** may reconnect to the other network devices in the media playback system via the wireless mesh network, enabling the network device **608** to transmit network connectivity results to the user device **602**.

(136) The user device **602** may then transmit a request to the network devices **608**, **610**, and **612** for their respective network connectivity test results. In turn, each of the network devices **608**, **610**, and **612** may transmit its respective network connectivity test results to the user device **602**. Upon receiving the network connectivity test results, the user device **602** may determine that the network device **608** was unable to connect to the other network devices in the media playback system via

the LAN, while the network devices **610** and **612** were able to connect to the other network devices in the media playback system via the LAN.

(137) The user device **602** may then communicate this determination to the user via a display on the user device **602**, indicating that a removal of the network device **606** would disrupt at least a portion of the currently arranged media playback system. The user device **602** may additionally communicate further troubleshooting steps or additional options to the user.

(138) FIG. **6B** depicts an alternative media playback system network topology **600** that includes the same user device **602**, router **604**, and network devices **606**, **608**, **610**, and **612**. However, as shown in FIG. **6B**, the network topology **600** includes the user device **602** and the network devices **610** and **612** connected to the other network devices in the media playback system via a wireless communication protocol, the network device **606** connected to the other network devices in the media playback system via a wired communication protocol, and the network device **608** connected to the other network devices in the media playback system via a network created by the network device **606**. Further, in FIG. **6B**, each of the network devices **606**, **608**, **610**, and **612** may be capable of connecting to the media playback device directly via the LAN.

(139) Similar to FIG. **6A**, in practice, applying the steps of the method flow diagram **500** described above with respect to FIG. **5** to the network topology **600** shown in FIG. **6B** may first include the user device **602**, after validating and transmitting the network credentials of the LAN, transmitting a command to the network device **606** to temporarily disconnect from the other network devices (e.g., the user device **602** and the network devices **608**, **610**, and **612**) in the media playback system via the wired communication protocol. As previously discussed, this command might be sent at an initialization time and may include a command for the network device **606** to, at a first time, temporarily disable a network interface that corresponds to the wired communication protocol. The command might also include a command to, after a delay, reenabling the network interface that corresponds to the wired communication protocol.

(140) The user device **602** may also transmit a command to each of the network devices **608**, **610** and **612** to, at a second time while the network device **606** is temporarily disconnected from the LAN, test its network connectivity with other network devices in the media playback system. In this example, the second time is after the first time associated with the network device **606** temporarily disconnecting from the other network devices in the media playback system but before a third time at which the network device **606** reconnects to the other network devices in the media playback system. In response to receiving the command to test its respective network connectivity, each of the network devices **608**, **610**, and **612** will attempt to query the user device **602** via the LAN. As previously mentioned, each of the network devices **608**, **610**, and **612** in the example shown in FIG. **6B** may be capable of independently connecting to the LAN and thus each of the network devices **608**, **610**, and **612** may query the user device **602** without issue.

(141) After the network device **606** reconnects, the user device **602** may then transmit a request to the network devices **608**, **610**, and **612** for their respective network connectivity test results. In turn, each of the network devices **608**, **610**, and **612** may transmit its respective network connectivity test results to the user device **602**. Upon receiving the network connectivity test results, the user device **602** may determine that each of the network devices **608**, **610**, and **612** was able to connect to the media playback system. In turn, the user device **602** may communicate this determination to the user via a display on the user device **602**. The user **602** may additionally communicate an indication that the user can remove the network device **606** from the media playback system without affecting performance of the rest of the media playback system.

(142) FIG. **6C** depicts another alternative media playback system network topology **600** that includes the user device **602**, router **604**, and network devices **606**, **608**, **610**, and **612**. However, the network device **606** may be an older playback device that is to be removed, while the network device **608**, **610**, and **612** may be newer playback devices. As such, the network topology **600** includes the user device **602** and the network devices **606**, **608**, **610** and **612** connected to the

media playback system via a wireless communication protocol. Further, in FIG. 6C, each of the network devices **606**, **608**, **610**, and **612** may be capable of connecting to the media playback device directly via the LAN.

(143) Similar to FIGS. 6A and 6B, in practice, applying the steps of the method flow diagram **500** described above with respect to FIG. 5 to the network topology **600** shown in FIG. 6C may first include the user device **602**, after validating and transmitting the network credentials of the LAN, transmitting a command to the network device **606** to temporarily disconnect from the other network devices (e.g., the user device **602** and the network devices **608**, **610**, and **612**) in the media playback system via the wireless communication protocol. As previously discussed, this command might be sent at an initialization time and may include a command for the network device **606** to, at a first time, temporarily disable a network interface that corresponds to the wireless communication protocol. The command might also include a command to, after a delay, reenabling the network interface that corresponds to the wireless communication protocol.

(144) The user device **602** may also transmit a command to each of the network devices **608**, **610** and **612** to, at a second time while the network device **606** is temporarily disconnected from the LAN, test its network connectivity with other network devices in the media playback system. In this example, the second time is after the first time associated with the network device **606** temporarily disconnecting from the other network devices in the media playback system but before a third time at which the network device **606** reconnects to the other network devices in the media playback system. In response to receiving the command to test its respective network connectivity, each of the network devices **608**, **610**, and **612** will attempt to query the user device **602** via the LAN. As previously mentioned, each of the network devices **608**, **610**, and **612** may be capable of connecting to the LAN and thus each of the network devices **608**, **610**, and **612** may query the user device **602** without issue.

(145) The user device **602** may then transmit a request to the network devices **608**, **610**, and **612** for their respective network connectivity test results. In turn, each of the network devices **608**, **610**, and **612** may transmit its respective network connectivity test results to the user device **602**. Upon receiving the network connectivity test results, the user device **602** may determine that each of the network devices **608**, **610**, and **612** was able to connect to the media playback system. In turn, the user device **602** may communicate this determination to the user via a display on the user device **602**. The user **602** may additionally communicate an indication that the user can remove the network device **606** from the media playback system without affecting performance of the rest of the media playback system.

(146) FIG. 6D depicts yet another alternative media playback system network topology **600** that includes the user device **602**, router **604**, and network devices **606**, **608**, **610**, and **612**. The network devices **606**, **608**, and **612** might be newer playback device, while the network device **610** might be an older playback device that is to be removed. The network topology **600** may include the user device **602** and the network devices **606** and **608** connected to the media playback system via a wireless communication protocol, the network device **610** connected to the media playback system via a wired communication protocol, and the network device **612** connected to a wireless mesh network created by the network device **610**. Further, in FIG. 6D, each of the network devices **606**, **608**, and **612** may be capable of connecting to the media playback device directly via the LAN, even though the network device **612** may not be close enough to the router **604** to do so.

(147) In practice, applying the steps of the method flow diagram **500** described above with respect to FIG. 5 to the network topology **600** shown in FIG. 6D may first include the user device **602**, after validating and transmitting the network credentials of the LAN, transmitting a command to the network device **610** to temporarily disconnect from the other network devices (e.g., the user device **602** and the network devices **606**, **608**, and **612**) in the media playback system via the wired communication protocol. As previously discussed, this command might be sent at an initialization time and may include a command for the network device **610** to, at a first time, temporarily disable

a network interface that corresponds to the wired communication protocol. The command might also include a command to later reenble the network interface that corresponds to the wired communication protocol.

(148) The user device **602** may then transmit a command to each of the network devices **606**, **608** and **612** to, at a second time while the network device **610** is temporarily disconnected from the LAN, test its network connectivity with other network devices in the media playback system. In this example, the second time is after the first time associated with the network device **610** temporarily disconnecting from the other network devices in the media playback system but before a third time at which the network device **610** reconnects to the other network devices in the media playback system. In response to receiving the command to test its respective network connectivity, each of the network devices **606**, **608**, and **612** will attempt to query the user device **602** via the LAN. As previously mentioned, each of the network devices **606**, **608**, and **612** may be capable of connecting to the LAN, but since the network device **612** is not within range of the router **604**, the network device **612** may be unable to query the user device **602**, while the network devices **606** and **608** may query the user device **602** without issue.

(149) After the network device **610** reconnects to the other network devices in the media playback system, the user device **602** may then transmit a request to the network devices **606**, **608**, and **612** for their respective network connectivity test results. In turn, each of the network devices **606**, **608**, and **612** may transmit its respective network connectivity test results to the user device **602**. Upon receiving the network connectivity test results, the user device **602** may determine that the network devices **606** and **608** were able to connect to the media playback system while the network device **610** was temporarily disconnected, while the **612** was unable to connect to the media playback system while the network device **610** was temporarily disconnected. In turn, the user device **602** may then communicate this determination to the user via a display on the user device **602**. The user **602** may additionally communicate further troubleshooting steps or additional options to the user.

(150) Further, it should be understood that the examples described above in connection to FIGS. **6A-6D** are for illustrative purposes and should not be considered as limiting. Accordingly, the examples provided are not the only ways to implement the message flow diagram **500** described with respect to FIG. **5** as other media playback system configurations and network topologies are contemplated herein. Further, it should be understood that combinations of various elements of each of the examples described in FIGS. **6A-6D** are also possible.

IV. Example Methods for Simulating Network Device Removal

(151) FIG. **7** shows an example flow diagram of a method **700** that can be implemented by a computing device, such as the controller device **104** described with respect to FIGS. **4A**, **4B**, and **4C**, the user device **502** described with respect to FIG. **5**, and the user device **602** described with respect to FIGS. **6A**, **6B**, **6C**, and **6D**.

(152) Various embodiments of the method **700** include one or more operations, functions, and actions illustrated by blocks **702**, **704**, and **706**. Although the blocks are illustrated in sequential order, these blocks may also be performed in parallel, and/or in a different order than the order disclosed and described herein. Also, the various blocks may be combined into fewer blocks, divided into additional blocks, and/or removed based upon a desired implementation.

(153) In addition, for the method **700** and other processes and methods disclosed herein, the flow diagram shows functionality and operation of one possible implementation of some embodiments. In this regard, each block may represent a module, a segment, or a portion of program code, which includes one or more instructions executable by one or more processors for implementing specific logical functions or steps in the process. The program code may be stored on any type of computer readable medium, for example, such as a storage device including a disk or hard drive. The computer readable medium may include non-transitory computer readable media, for example, such as tangible, non-transitory computer-readable media that stores data for short periods of time like register memory, processor cache, and Random Access Memory (RAM). The computer

readable medium may also include non-transitory media, such as secondary or persistent long-term storage, like read only memory (ROM), optical or magnetic disks, compact-disc read only memory (CD-ROM), for example. The computer readable media may also be any other volatile or non-volatile storage systems. The computer readable medium may be considered a computer readable storage medium, for example, or a tangible storage device. In addition, for the method **700** and other processes and methods disclosed herein, each block in FIG. **7** may represent circuitry that is wired to perform the specific logical functions in the process.

(154) The method **700** may begin at block **702** with a computing device, while operating as a control device of a networked media playback system that includes a first set of network devices and a second set of network devices, transmitting, via a network interface, a first message to each network device in the first set of network devices. The first message may include a command for the network device in the first set of network devices to, at a first time, temporarily disconnect from the computing device via a particular communication protocol, as discussed above in relation to FIGS. **5** and **6A-6D**. For example, the command may be for each network device in the first set of network devices to disconnect following a first delay. In this regard, the first message may be transmitted at an initialization time that is before the first time.

(155) At block **704**, the method **700** may involve transmitting, via the network interface, a second message to each network device in the second set of network devices that includes a command for the network device in the second set of network devices to, at a second time, test connectivity with the computing device. For example, the command may be for each network device in the second set of network devices to test its connectivity with the computing device following a second delay. In this regard, the second delay may be greater than the first delay, as shown in FIG. **5** and discussed above.

(156) Returning to block **702**, the first message may further include a command for each network device in the first set of network devices to reestablish connection with the computing device via the particular communication protocol at a third time that follows the second time. In this regard, the command may be a command to reestablish connection with the computing device via the particular communication protocol following a third delay, which may be greater than the second delay, as shown in FIG. **5**. Accordingly, to increase the likelihood of accurate connectivity test results at block **704**, the second message that the computing device transmits to the second set of network devices may also include a command for the network device in the second set of network devices to, before the third time, cease testing connectivity with the computing device.

(157) At block **706**, the method **700** may involve the computing device, after the second time, determining whether each network device in the second set of network devices is connected to the computing device. In some examples, determining whether each network device in the second set of network devices is connected to the computing device may involve determining that one or more network devices in the of the second set of network devices failed to connect to the computing device while the first set of network devices were temporarily disconnected from the computing device, as described above with respect to the examples shown in FIGS. **6A** and **6D**. After determining whether each network device in the second set of network devices was connected to the computing device during the connectivity test, the computing device may display an indication of whether each network device in the second set of network devices was connected to the computing device.

IV. Conclusion

(158) The description above discloses, among other things, various example systems, methods, apparatus, and articles of manufacture including, among other components, firmware and/or software executed on hardware. It is understood that such examples are merely illustrative and should not be considered as limiting. For example, it is contemplated that any or all of the firmware, hardware, and/or software aspects or components can be embodied exclusively in hardware, exclusively in software, exclusively in firmware, or in any combination of hardware,

software, and/or firmware. Accordingly, the examples provided are not the only way(s) to implement such systems, methods, apparatus, and/or articles of manufacture.

(159) Further, the examples described herein may be employed in systems separate and apart from media playback systems such as any Internet of Things (IoT) system comprising an IoT device. An IoT device may be, for example, a device designed to perform one or more specific tasks (e.g., making coffee, reheating food, locking a door, providing power to another device, playing music) based on information received via a network (e.g., a WAN such as the Internet). Example IoT devices include a smart thermostat, a smart doorbell, a smart lock (e.g., a smart door lock), a smart outlet, a smart light, a smart vacuum, a smart camera, a smart television, a smart kitchen appliance (e.g., a smart oven, a smart coffee maker, a smart microwave, and a smart refrigerator), a smart home fixture (e.g., a smart faucet, a smart showerhead, smart blinds, and a smart toilet), and a smart speaker (including the network accessible and/or voice-enabled playback devices described above). These IoT systems may also comprise one or more devices that communicate with the IoT device via one or more networks such as one or more cloud servers (e.g., that communicate with the IoT device over a WAN) and/or one or more computing devices (e.g., that communicate with the IoT device over a LAN and/or a PAN). Thus, the examples described herein are not limited to media playback systems.

(160) It should be appreciated that references to transmitting information to particular components, devices, and/or systems herein should be understood to include transmitting information (e.g., messages, requests, responses) indirectly or directly to the particular components, devices, and/or systems. Thus, the information being transmitted to the particular components, devices, and/or systems may pass through any number of intermediary components, devices, and/or systems prior to reaching its destination. For example, a control device may transmit information to a playback device by first transmitting the information to a computing system that, in turn, transmits the information to the playback device. Further, modifications may be made to the information by the intermediary components, devices, and/or systems. For example, intermediary components, devices, and/or systems may modify a portion of the information, reformat the information, and/or incorporate additional information.

(161) Similarly, references to receiving information from particular components, devices, and/or systems herein should be understood to include receiving information (e.g., messages, requests, responses) indirectly or directly from the particular components, devices, and/or systems. Thus, the information being received from the particular components, devices, and/or systems may pass through any number of intermediary components, devices, and/or systems prior to being received. For example, a control device may receive information from a playback device indirectly by receiving information from a cloud server that originated from the playback device. Further, modifications may be made to the information by the intermediary components, devices, and/or systems. For example, intermediary components, devices, and/or systems may modify a portion of the information, reformat the information, and/or incorporate additional information.

(162) The specification is presented largely in terms of illustrative environments, systems, procedures, steps, logic blocks, processing, and other symbolic representations that directly or indirectly resemble the operations of data processing devices coupled to networks. These process descriptions and representations are typically used by those skilled in the art to most effectively convey the substance of their work to others skilled in the art. Numerous specific details are set forth to provide a thorough understanding of the present disclosure. However, it is understood to those skilled in the art that certain embodiments of the present disclosure can be practiced without certain, specific details. In other instances, well known methods, procedures, components, and circuitry have not been described in detail to avoid unnecessarily obscuring aspects of the embodiments. Accordingly, the scope of the present disclosure is defined by the appended claims rather than the forgoing description of embodiments.

(163) When any of the appended claims are read to cover a purely software and/or firmware

implementation, at least one of the elements in at least one example is hereby expressly defined to include a tangible, non-transitory medium such as a memory, DVD, CD, Blu-ray, and so on, storing the software and/or firmware.

Claims

1. A computing device comprising: at least one processor; a user interface; at least one non-transitory computer-readable medium; and program instructions stored on the at least one non-transitory computer-readable medium that are executable by the at least one processor such that the computing device is configured to: operate as a control device of a plurality of networked media playback systems comprising at least a first networked media playback system; transmit, to a first network device of the first networked media playback system that is configured to communicate with the computing device via a particular communication protocol, a first command to, at a first time, simulate removal of the first network device from the first networked media playback system by temporarily disconnecting from the computing device via the particular communication protocol; transmit, to a second network device of the first networked media playback system, a second command to, at a second time while the first network device is simulating removal of the first network device from the first networked media playback system by temporarily disconnecting from the computing device, test connectivity of the second network device with the computing device; after the second time, obtain network connectivity test results for the second network device of the first networked media playback system while the first network device was simulating removal of the first network device from the networked media playback system by temporarily disconnecting from the computing device; and display, via the user interface, information about network connectivity test results for the second network device of the first networked media playback system.
2. The computing device of claim 1, wherein the control device is a remote control device that is configured to communicate with network devices of the first networked media playback system via a wide area network (WAN).
3. The computing device of claim 1, wherein the first command further comprises a command for the first network device to, at a third time after the second time, reestablish connection with the computing device via the particular communication protocol.
4. The computing device of claim 3, wherein the program instructions that are executable by the at least one processor such that the computing device is configured to transmit the first command comprise program instructions that are executable by the at least one processor such that the computing device is configured to transmit the first command at an initialization time before the first time.
5. The computing device of claim 1, wherein the network connectivity test results for the second network device indicate that the second network device failed to establish a connection with the computing device while the first network device was disconnected from the computing device, and wherein the information about the network connectivity test results for the second network device comprises an indication of a connectivity failure of the second network device.
6. The computing device of claim 1, wherein the information about the network connectivity test results for the second network device further comprises information for troubleshooting connectivity of the second network device.
7. The computing device of claim 1, further comprising program instructions stored on the at least one non-transitory computer-readable medium that are executable by the at least one processor such that the computing device is configured to: before transmitting the first command, receive, via the user interface, a request to test connectivity of one or more network devices of the first networked media playback system.
8. The computing device of claim 1, further comprising program instructions stored on the at least

one non-transitory computer-readable medium that are executable by the at least one processor such that the computing device is configured to: receive, via the user interface, a request to test connectivity of one or more network devices of a second networked media playback system of the plurality of networked media playback systems; transmit, to a third network device of the second networked media playback system that is configured to communicate with the computing device via the particular communication protocol, a first third command to, at a third time, simulate removal of the third network device from the second networked media playback system by temporarily disconnecting from the computing device via the particular communication protocol; transmit, to a fourth network device of the second networked media playback system, a fourth command to, at a fourth time while the third network device of the second networked media playback system is disconnected from the computing device, test respective connectivity of the fourth network device with the computing device; after the fourth time, obtain network connectivity test results for the fourth network device of the second networked media playback system while the third network device of the second networked media playback system was disconnected from the computing device; and display, via the user interface, information about network connectivity test results for the fourth network device of the second networked media playback system.

9. At least one non-transitory computer-readable medium, wherein the at least one non-transitory computer-readable medium is provisioned with program instructions that, when executed by at least one processor, cause a computing device to: operate as a control device of a plurality of networked media playback systems comprising at least a first networked media playback system; transmit, to a first network device of the first networked media playback system that is configured to communicate with the computing device via a particular communication protocol, a first command to, at a first time, simulate removal of the first network device from the first networked media playback system by temporarily disconnecting from the computing device via the particular communication protocol; transmit, to a second network device of the first networked media playback system, a second command to, at a second time while the first network device is simulating removal of the first network device from the networked media playback system by temporarily disconnecting from the computing device, test connectivity of the second network device with the computing device; after the second time, obtain network connectivity test results for the second network device of the first networked media playback system while the first network device was simulating removal of the first network device from the networked media playback system by temporarily disconnecting from the computing device; and display, via a user interface of the computing device, information about network connectivity test results for the second network device of the first networked media playback system.

10. The at least one non-transitory computer-readable medium of claim 9, wherein the control device is a remote control device that is configured to communicate with network devices of the first networked media playback system via a wide area network (WAN).

11. The at least one non-transitory computer-readable medium of claim 9, wherein the first command further comprises a command for the first network device to, at a third time after the second time, reestablish connection with the computing device via the particular communication protocol.

12. The at least one non-transitory computer-readable medium of claim 11, wherein the program instructions that, when executed by at least one processor, cause the computing device to transmit the first command comprise program instructions that, when executed by at least one processor, cause the computing device to transmit the first command at an initialization time before the first time.

13. The at least one non-transitory computer-readable medium of claim 9, wherein the network connectivity test results for the second network device indicate that the second network device failed to establish a connection with the computing device while the first network device was

disconnected from the computing device, and wherein the information about the network connectivity test results for the second network device comprises an indication of a connectivity failure of the second network device.

14. The at least one non-transitory computer-readable medium of claim 9, wherein the information about the network connectivity test results for the second network device further comprises information for troubleshooting the connectivity of the second network device.

15. The at least one non-transitory computer-readable medium of claim 9, wherein the at least one non-transitory computer-readable medium is also provisioned with program instructions that, when executed by at least one processor, cause the computing device to: before transmitting the first command, receive, via the user interface of the computing device, a request to test connectivity of one or more network devices of the first networked media playback system.

16. The at least one non-transitory computer-readable medium of claim 9, wherein the at least one non-transitory computer-readable medium is also provisioned with program instructions that, when executed by at least one processor, cause the computing device to: receive, via the user interface, a request to test connectivity of one or more network devices of a second networked media playback system of the plurality of networked media playback systems; transmit, to a third network device of the second networked media playback system that is configured to communicate with the computing device via the particular communication protocol, a first third command to, at a third time, simulate removal of the third network device from the second networked media playback system by temporarily disconnecting from the computing device ever via the particular communication protocol; transmit, to a fourth network device of the second networked media playback system, a fourth command to, at a fourth time while the third network device of the second networked media playback system is disconnected from the computing device, test respective connectivity of the fourth network device with the computing device; after the fourth time, obtain network connectivity test results for the fourth network device of the second networked media playback system while the third network device of the second networked media playback system was disconnected from the computing device; and display, via the user interface, information about network connectivity test results for the fourth network device of the second networked media playback system.

17. A method carried out by a computing device, the method comprising: operating as a control device of a plurality of networked media playback systems comprising at least a first networked media playback system; transmitting, to a first network device of the first networked media playback system that is configured to communicate with the computing device via a particular communication protocol, a first command to, at a first time, simulate removal of the first network device from the first networked media playback system by temporarily disconnecting from the computing device via the particular communication protocol; transmitting, to a second network device of the first networked media playback system, a second command to, at a second time while the first network device is simulating removal of the first network device from the networked media playback system by temporarily disconnecting from the computing device, test connectivity of the second network device with the computing device; after the second time, obtaining network connectivity test results for the second network device of the first networked media playback system while the first network device was simulating removal of the first network device from the networked media playback system by temporarily disconnecting from the computing device; and displaying, via a user interface of the computing device, information about network connectivity test results for the second network device of the first networked media playback system.

18. The method of claim 17, wherein the control device is a remote control device that is configured to communicate with network devices of the first networked media playback system via a wide area network (WAN).

19. The method of claim 17, wherein the first command further comprises a command for the first network device to, at a third time after the second time, reestablish connection with the computing

device via the particular communication protocol.

20. The method of claim 17, wherein the network connectivity test results for the second network device indicate that the second network device failed to establish a connection with the computing device while the first network device was disconnected from the computing device, and wherein the information about the network connectivity test results for the second network device comprises an indication of a connectivity failure of the second network device.
